

Kathryn Marek

Madison, WI

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Recent B.S. graduate with 2+ years of research and data experience in ArcGIS Pro and Python, seeking a geospatial analyst role focused on environmental and water-resource mapping and analysis.

Education

University of Wisconsin–Madison

Sep 2022 – May 2026

*B.S. in Atmospheric and Oceanic Science; Certificates in **Geospatial Data Science**, Mathematics, and Environmental Studies*

Relevant coursework: GIS & Spatial Analysis, Spatial Database Design, Remote Sensing, Statistics, Data Science

GIS & Technical Skills

GIS Software: ArcGIS Pro, ArcGIS Online, Field Maps, Survey123, Adobe Illustrator

Analysis & Cartography: spatial analysis, map production, geospatial data management, coordinate systems, projections

Programming & Databases: Python, R, SQL, PostgreSQL/pgAdmin4, Git/GitHub

Data Management: QA/QC procedures, metadata standards, data request fulfillment, satellite & gridded data

Field & Communication: GPS field data collection, environmental sampling, technical writing/reporting, scientific presentations, MS Office (Word, Excel, Teams), valid driver's license

Relevant Experience

NASA Ames Research Center – EarthRISE Developer's Academy

Sep 2026 – Present

Early Career Researcher

Mountain View, CA

- Integrate GRACE and GRACE-FO satellite observations with hydrologic modeling and ancillary datasets to estimate groundwater availability in the High Cascades of Oregon and California
- Analyze results in Python and ArcGIS Pro and build deliverables and end products that support water-resource decision-making for stakeholders across the western U.S.

Antarctic Meteorological Research and Data Center

Feb 2024 – May 2026

Data Intern

Madison, WI

- Quality-controlled large scientific datasets, maintaining metadata standards and documented QA/QC procedures to ensure accuracy and institutional compliance
- Produced maps and geospatial products in ArcGIS Pro for publications, presentations, and public outreach requests

SETI Institute

Jun 2025 – Aug 2025

Atmospheric Science Research Intern

Mountain View, CA

- Applied GIS and Python to analyze dust devil activity across space and time, building geospatial datasets and visualizations that communicated findings to scientific audiences
- Presented findings at AGU 2025 and AMS 2026; co-author of a manuscript in preparation

Northeast Partnership in Atmospheric and Related Sciences

Jun 2024 – Aug 2024

Environmental Science Intern

Geneva, NY

- Collected sediment cores and water quality samples in the field to support baseline environmental assessment and reporting for the Seneca Lake watershed
- Used R to reconstruct long-term hydrologic variability and evaluate historical floods, informing infrastructure and water resource planning

Relevant Projects

Billion Dollar Disasters: Climate Variability and Income Across U.S. States

Spring 2026

Spatial Database Design & Analysis

GEOG574, UW-Madison

- Designed a relational PostgreSQL database (ER diagram, schema, foreign keys) integrating NOAA climate, disaster, and USDA economic datasets
- Wrote SQL queries (CTEs, multi-table joins, aggregations) and produced ArcGIS choropleth maps to communicate disaster costs and income disparities

Climatological Assessment of Ross Island, Antarctica

Fall 2025

Statistical Analysis & Geospatial Visualization

AOS573, UW-Madison

- Built a Python pipeline (pandas, NumPy) to clean, quality-control, and analyze 20 years of multi-station time series data, visualizing relationships across stations and time

Leadership

American Meteorological Society, UW-Madison Chapter

Sep 2023 – May 2026

President 2025–2026, Junior Officer 2024–2025

University of Wisconsin–Madison

- Led board meetings and coordinated events for a 50+ member organization; managed outreach with guest speakers and external organizations